

UNIT-III

Introduction to Analysis and Analytics, Tools and Environment, Application of Modelling in Business, Data Modelling Techniques, Missing Imputations, Need for Business Modelling.

Introduction to Analysis

Data analysis is a key part of data analytics. It involves scrutinizing existing data to gain insights and draw conclusions. While data analysis focuses on looking at data on hand to identify patterns and insights, data analytics uses a broader range of techniques, including predictive modeling and machine learning, to not only interpret past data but also forecast future trends.

Descriptive Analysis

Descriptive data analysis focuses on summarizing and describing data to gain an understanding of patterns, trends, and relationships. It tells you what already happened instead of predicting future outcomes.

Diagnostic Analysis

This type involves digging deeper into the data to understand why certain trends or patterns are occurring and identifying potential causes. It seeks to answer why something happened.

Predictive Analysis

Using statistical techniques and machine learning algorithms, this type aims to make predictions about future outcomes based on historical data. It works towards identifying what can happen.

Prescriptive Analysis

Building upon predictive analysis, prescriptive analysis provides recommendations for actions that can be taken to achieve desired outcomes. It aims to answer what can be done.

Data Analysis Techniques

Regression Analysis

Regression analysis is a statistical technique that examines the relationship between two or more variables to determine how they are related and predict future outcomes. It helps identify patterns and trends in data, making it useful for predictive analysis.

Cluster Analysis

Cluster analysis involves grouping data into similar clusters based on shared characteristics, helping identify patterns and segments within larger datasets. It is often used in customer segmentation, marketing research, and social sciences.

Time Series Analysis

Time series analysis is a statistical technique used to analyze trends over time. It involves examining data collected at regular intervals to identify patterns and forecast future trends.

Text Mining

Text mining is the process of analyzing large amounts of unstructured text data to uncover insights and patterns that can inform decision-making. It often involves using natural language processing techniques to extract valuable information from text documents, such as customer feedback, social media posts, or online reviews.

Introduction to Analytics

Analytics is the science of analysis wherein we apply methods of statistics, data mining and computer technology for doing the analysis, whereas analysis is the process wherein the complex data available is being broken down into simpler forms which provides more compact and better data for understanding.

There are major four stages in any analytics work-

Stage 1: Descriptive Analytics

Where data or information is gathered and summarized upon. This stage usually caters to questions like “How many students dropped out last year?”

Stage 2: Diagnostic Analytics

Where data is analysed, and insights are generated which help in answering the question in the first stage. Here the question that comes up will be “Why has the dropout rate increased in the last one year?”

Stage 3: Predictive analysis

With the help of the analyses done in the previous two stages, this stage tries to answer unforeseen phenomena like “Which students are most likely to drop out?”

Stage 4: Prescriptive Analytics

Finally, the last stage tries to analyze the type action required to be taken to support or avoid the unforeseen phenomena predicted in the previous stage. In this scenario, prescriptive analytics has queries like “Which students should I target to keep from dropping out?”

Analytics is a journey that involves a combination of potential skills, advanced technologies, applications, and processes used by firm to gain business insights from data and statistics. This is done to perform business planning.

Reporting Vs Analytics:

Reporting is presenting result of data analysis and Analytics is process or systems involved in analysis of data to obtain a desired output.

Introduction to Tools and Environment

Analytics is now days used in all the fields ranging from Medical Science to Aero science to Government Activities.

Data Science and Analytics are used by Manufacturing companies as well as Real Estate firms to develop their business and solve various issues by the help of historical data base.

Tools are the softwares that can be used for Analytics like SAS or R. While techniques are the procedures to be followed to reach up to a solution.

1. Excel

With a variety of powerful features such as form creation, PivotTable, VBA, etc., Excel's system is so large that no analytics tool can surpass it, ensuring that people can analyze data according to their needs.

2. BI tools

Business intelligence is born for data analysis, and it is born at a very high starting point. The goal is to shorten the time from business data to business decisions and use data to influence decisions.

Tableau

The core essence of Tableau is actually the PivotTable and PivotChart of Excel. It can be said that Tableau is keenly aware of this feature of Excel. It entered the BI market earlier and carried forward this core value.

Power BI

The advantage of Power BI lies in its business model and data analysis capabilities.

Power BI was previously a plug-in for Excel, and its development was not ideal. So it got out of Excel and developed into a BI tool. As a latecomer, Power BI has iterative updates every month and catches up very fast.

Fine Report

What makes FineReport unique is that its self-service data analysis is very suitable for business users. With a simple drag and drop operation, you can design various styles of reports with FineReport and easily build a data decision analysis system.

3. R & Python

R and Python are the third type of tools, although software like Excel and BI tools have been designed with the utmost effort to consider the most application scenarios of data analysis.

For example, R and Python are the indispensable tools for data scientists. From a professional perspective, they are definitely powerful than Excel and BI tools.

Application of Modelling in Business

A statistical model embodies a set of assumptions concerning the generation of the observed data, and similar data from a larger population.

A model represents, often in considerably idealized form, the data-generating process. □ Signal processing is an enabling technology that encompasses the fundamental theory, applications, algorithms, and implementations of processing or transferring information contained in many different physical, symbolic, or abstract formats broadly designated as signals.

It uses mathematical, statistical, computational, heuristic, and linguistic representations, formalisms, and techniques for representation, modeling, analysis, synthesis, discovery, recovery, sensing, acquisition, extraction, learning, security, or forensics.

In manufacturing statistical models are used to define Warranty policies, solving various conveyor related issues, Statistical Process Control etc.

What is Data Modelling?

Data Modelling is a set of tools and techniques used to understand and analyse how an organisation should collect, update, and store data. It is a critical skill for the business analyst who is involved with discovering, analysing, and specifying changes to how software systems create and maintain information.

Or

Data modelling is the process of creating a data model for the data to be stored in a Database. This data model is a conceptual representation of Data objects, the associations between different data objects and the rules. Data modelling helps in the visual representation of data and enforces business rules, regulatory compliances, and government policies on the

data. Data Models ensure consistency in naming conventions, default values, semantics, security while ensuring quality of the data.

What does a Data Modeller do?

They create an entity relationship diagram to visualise relationships between key business concepts.

They create a conceptual-level data dictionary to communicate data requirements that are important to business stakeholders.

They create a data map to resolve potential data issues for a data migration or integration project.

A data modeller would not necessarily query or manipulate data or become involved in designing or implementing databases or data repositories.

Why use Data Model?

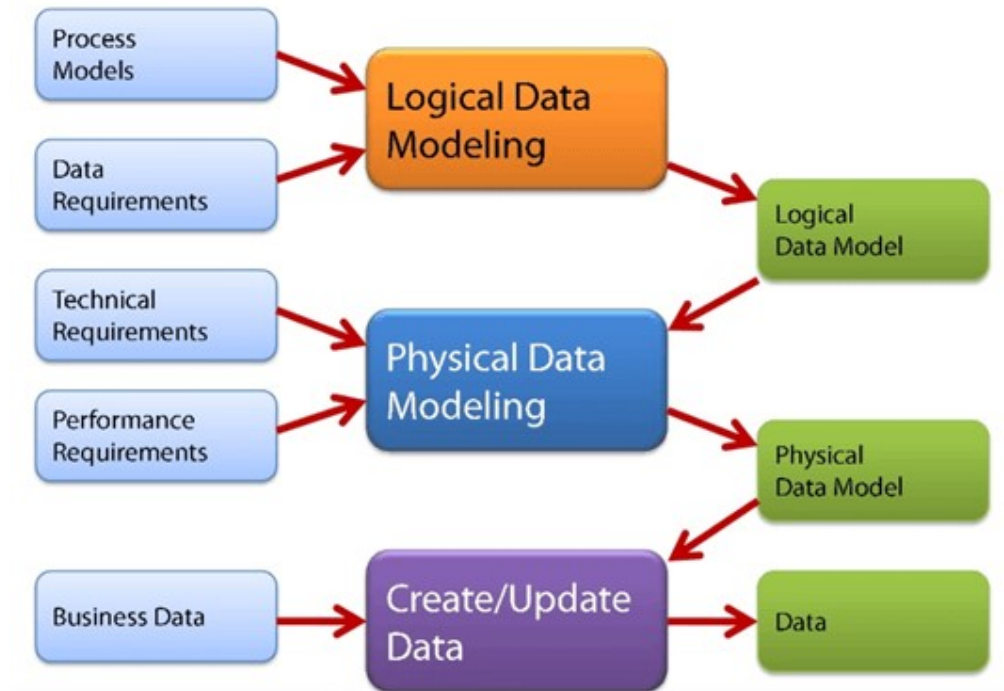
The primary goals of using data model are:

- Ensures that all data objects required by the database are accurately represented. Omission of data will lead to creation of faulty reports and produce incorrect results.
- A data model helps design the database at the conceptual, physical and logical levels.
- Data Model structure helps to define the relational tables, primary and foreign keys and stored procedures.
- It provides a clear picture of the base data and can be used by database developers to create a physical database.
- It is also helpful to identify missing and redundant data.

TYPES OF DATA MODELS

There are mainly three different types of data models:

1. **Conceptual:** This Data Model defines **WHAT** the system contains. This model is typically created by Business stakeholders and Data Architects. The purpose is to organize, scope and define business concepts and rules.
2. **Logical:** Defines **HOW** the system should be implemented regardless of the DBMS. This model is typically created by Data Architects and Business Analysts. The purpose is to develop technical map of rules and data structures.
3. **Physical:** This Data Model describes **HOW** the system will be implemented using a specific DBMS system. This model is typically created by DBA and developers. The purpose is actual implementation of the database.



Conceptual Model

The main aim of this model is to establish the entities, their attributes, and their relationships. In this Data modelling level, there is hardly any detail available of the actual Database structure.

The 3 basic tenants of Data Model are

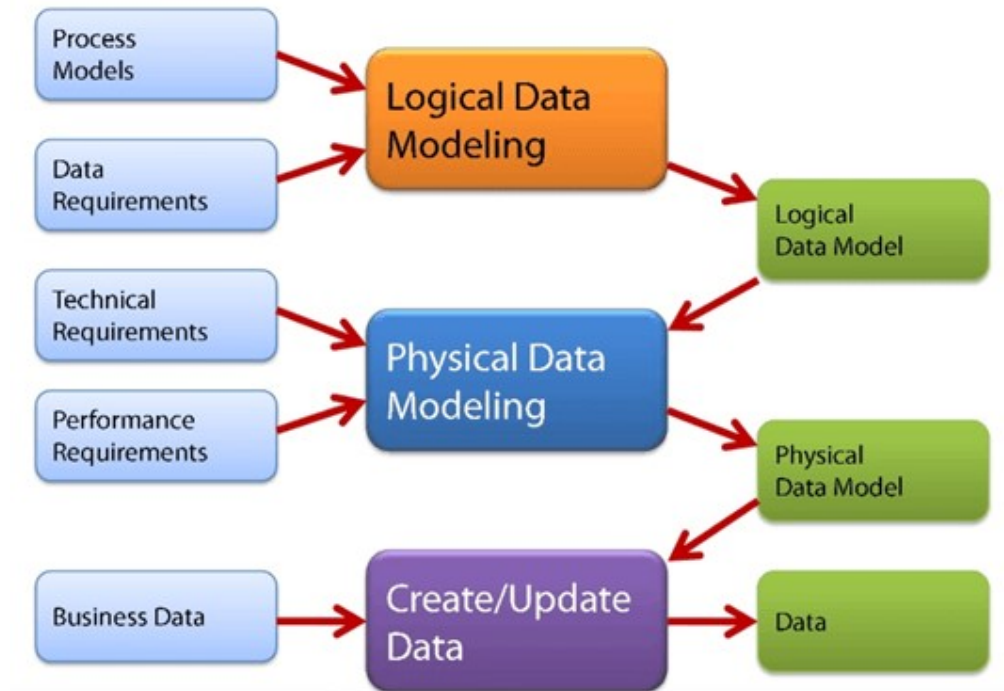
Entity: A real-world thing

Attribute: Characteristics or properties of an entity

Relationship: Dependency or association between two entities

For example:

- Customer and Product are two entities. Customer number and name are attributes of the Customer entity
- Product name and price are attributes of product entity
- Sale is the relationship between the customer and product



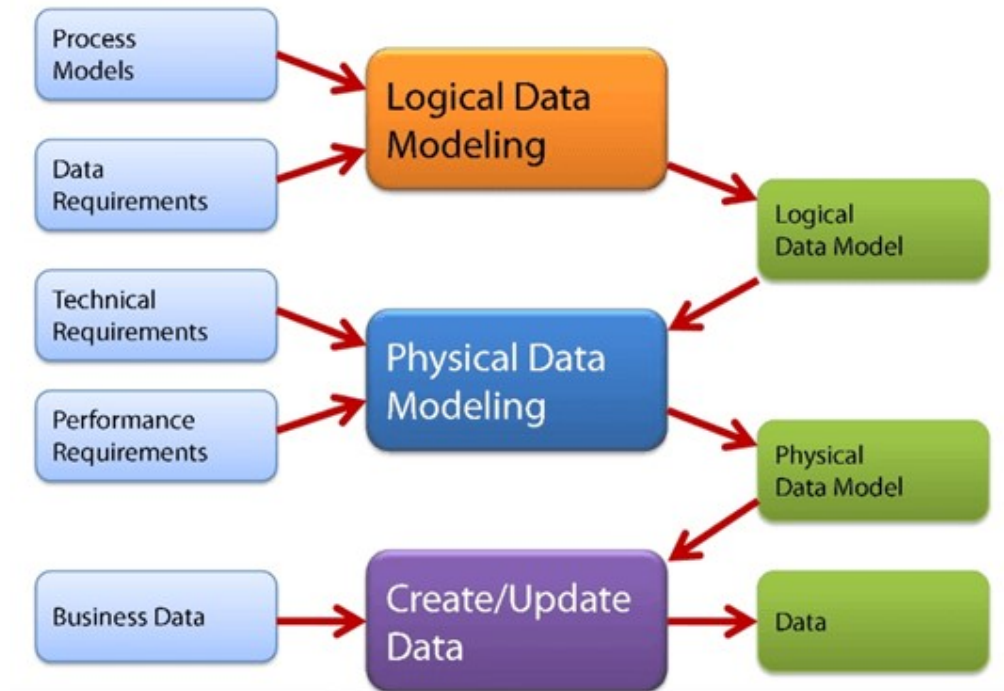
Characteristics of a conceptual data model

- Offers Organisation-wide coverage of the business concepts.
- This type of Data Models are designed and developed for a business audience.
- The conceptual model is developed independently of hardware specifications like data storage capacity, location or software specifications like DBMS vendor and technology. The focus is to represent data as a user will see it in the "real world."

Conceptual data models known as Domain models create a common vocabulary for all stakeholders by establishing basic concepts and scope.

Logical Data Model

Logical data models add further information to the conceptual model elements. It defines the structure of the data elements and set the relationships between them.



The advantage of the Logical data model is to provide a foundation to form the base for the Physical model. However, the modelling structure remains generic.

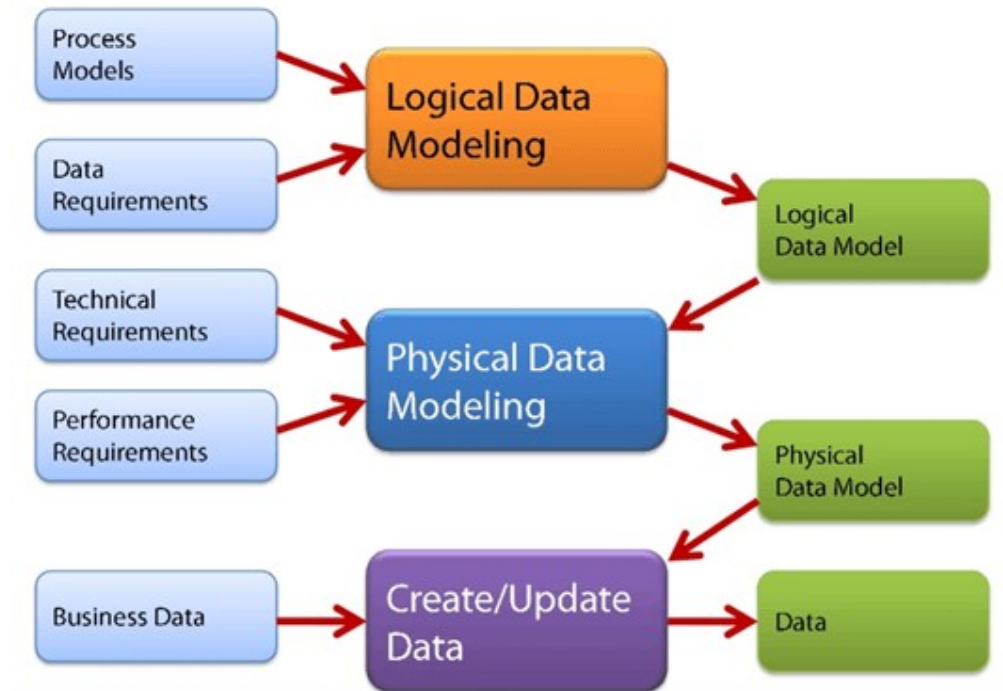
At this Data Modelling level, no primary or secondary key is defined. At this Data modeling level, you need to verify and adjust the connector details that were set earlier for relationships.

Characteristics of a Logical data model

- Describes data needs for a single project but could integrate with other logical data models based on the scope of the project.
- Designed and developed independently from the DBMS.
- Data attributes will have datatypes with exact precisions and length.
- Normalization processes to the model is applied typically till 3NF.

Physical Data Model

A Physical Data Model describes the database specific implementation of the data model. It offers an abstraction of the database and helps generate schema. This is because of the richness of meta-data offered by a Physical Data Model.



This type of Data model also helps to visualize database structure. It helps to model database columns keys, constraints, indexes, triggers, and other RDBMS features.

Characteristics of a physical data model:

- The physical data model describes data need for a single project or application though it maybe integrated with other physical data models based on project scope.
- Data Model contains relationships between tables that which addresses cardinality and nullability of the relationships.
- Developed for a specific version of a DBMS, location, data storage or technology to be used in the project.
- Columns should have exact datatypes, lengths assigned and default values.
- Primary and Foreign keys, views, indexes, access profiles, and authorizations, etc. are defined.

ADVANTAGES AND DISADVANTAGES OF DATA MODEL:

Advantages of Data model:

- The main goal of a designing data model is to make certain that data objects offered by the functional team are represented accurately.
- The data model should be detailed enough to be used for building the physical database.
- The information in the data model can be used for defining the relationship between tables, primary and foreign keys, and stored procedures.
- Data Model helps business to communicate the within and across organizations.
- Data model helps to documents data mappings in ETL process
- Help to recognize correct sources of data to populate the model

Disadvantages of Data model:

- To develop Data model one should know physical data stored characteristics.
- This is a navigational system produces complex application development, management. Thus, it requires a knowledge of the biographical truth.
- Even smaller change made in structure requires modification in the entire application.
- There is no set data manipulation language in DBMS.

DATA MODELLING TECHNIQUES

There are three basic data modelling techniques

1. Entity Relationship Diagrams
2. UML Class Diagrams
3. Data Dictionary

1. Entity Relationship Diagrams

Also referred to as **ER diagrams** or **ERDs**. Entity-Relationship modeling is a default technique for modeling and the design of relational (traditional) databases. In this notation architect identifies:

- **Entities** representing objects (or tables in relational database),
- **Attributes** of entities including **data type**,
- **Relationships** between entities/objects (or foreign keys in a database).

ERDs work well to design a relational (classic) database, Excel databases or CSV files. Any kind of tabular data work well for visualization of database schemas and communication of top-level view of data.

2. UML Class Diagrams

UML (Unified Modeling Language) is a standardized family of notations for modeling and design of information systems. It was derived from various existing notations to provide a standard for software engineering. It comprises of several different diagrams representing different aspect of the system, and one of them being a **Class Diagram** that can be used for data modeling. Class diagrams are equivalent of ERDs in relational world and are mostly used to design classes in object-oriented programming languages (such as Java or C#).

In class diagrams architects define:

- **Classes** (equivalent of entity in relational world),
- **Attributes** of a class (same as in an ERD) including **data type**,
- **Methods** associated to specific class, representing its behavior (in relational world those would be stored procedures),
- **Relationships** grouped into two categories:
 - **Relationships between objects** (instances of Classes) differentiated into Dependency, Association, Aggregation and Composition (equivalent to relationships in an ERD),

- **Relationships between classes** of two kinds Generalization/Inheritance and Realization/Implementation (this has no equivalent in relational world).

You can use class diagrams to design a tabular data (such as in RDBMS), but were designed and are used mostly for object-oriented programs (such as Java or C#).

3. Data Dictionary

Data dictionaries are a tabular definition/representation of data assets. Data dictionary is an inventory of data sets/tables with the list of their attributes/columns.

Core data dictionary elements:

- List of data sets/tables
- List of attributes/columns of each table with data type.

Optional data dictionary elements:

- Item descriptions
- Relationships between tables/columns,
- Additional constraints, such as uniqueness, default values, value constraints or calculated columns.

Data dictionary is suitable as detailed specification of data assets and can be supplemented with ER diagrams, as both serve slightly different purpose.

MISSING IMPUTATIONS:

In statistics, **imputation** is the process of replacing missing data with substituted values. When substituting for a data point, it is known as "unit imputation"; when substituting for a component of a data point, it is known as "item imputation". There are three main problems that missing data causes: missing data can introduce a substantial amount of bias, make the handling and analysis of the data more arduous, and create reductions in efficiency. Because missing data can create problems for analyzing data, imputation is seen as a way to avoid pitfalls involved with list wise deletion of cases that have missing values. That is to say, when one or more values are missing for a case, most statistical packages default to discarding any case that has a missing value, which may introduce bias or affect the representativeness of the results.

Missing data is a common problem in practical data analysis. They are simply observations that we intend to make but did not. In datasets, missing values could be represented as '?', 'nan', 'N/A', blank cell, or sometimes '-999', 'inf', '-inf'. The following provides an introduction of missing data and describe some basic methods on how to handle them.

Imputation simply means that we replace the missing values with some guessed/estimated ones.

Mean, median, mode imputation

A simple guess of a missing value is the mean, median, or mode (most frequently appeared value) of that variable.

Regression imputation

Mean, median or mode imputation only look at the distribution of the values of the variable with missing entries. If we know there is a correlation between the missing value and other variables, we can often get better guesses by regressing the missing variable on other variables.

K-nearest neighbour (KNN) imputation

Besides model-based imputation like regression imputation, neighbour-based imputation can also be used. **K-nearest neighbour (KNN) imputation** is an example of neighbour-based imputation. For a discrete variable, **KNN imputer** uses the most frequent value among the k nearest neighbours and, for a continuous variable, use the mean or mode.

Depending on the nature of the data or data type, some other imputation methods may be more appropriate.

For example, for longitudinal data, such as patients' weights over a period of visits, it might make sense to use last valid observation to fill the NA's. This is known as **Last observation carried forward (LOCF)**

In other cases, for instance, if we are dealing with time-series data, it might make senses to use **interpolation** of observed values before and after a timestamp for missing values.

Multiple imputations

The Mean, median, mode imputation, regression imputation, stochastic regression imputation, KNN imputer are all methods that create a single replacement value for each missing entry. **Multiple Imputation (MI)**, rather than a different method, is more like a general approach/framework of doing the imputation procedure multiple times to create different plausible imputed datasets. The key motivation to use MI is that a single imputation cannot reflect sampling variability from both sample data and missing values.

Methods to Handle Missing Data

Data can be missing in the following ways:

- **Missing Completely At Random (MCAR):** When missing values are randomly distributed across all observations, then we consider the data to be missing completely at random. A quick check for this is to compare two parts of data – one with missing observations and the other without missing observations. On a t-test, if we do not find any difference in means between the two samples of data, we can assume the data to be MCAR.
- **Missing At Random (MAR):** The key difference between MCAR and MAR is that under MAR the data is not missing randomly across all observations, but is missing randomly only within sub-samples of data. For example, if high school GPA data is missing randomly across all schools in a district, that data will be considered MCAR. However, if data is randomly missing for students in specific schools of the district, then the data is MAR.
- **Not Missing At Random (NMAR):** When the missing data has a structure to it, we cannot treat it as missing at random. In the above example, if the data was missing for all students from specific schools, then the data cannot be treated as MAR.

The Role of Data Modeling in Business

Data modeling and business analysis are two closely related concepts in the field of business management and information technology. They're both essential for understanding, organizing, and optimizing business processes and data resources.

However, it is data modeling that plays a key role in business analysis in many ways - for instance:

- Data modeling helps business analysts understand and document the data requirements of an organization, as well as gain insights into what kind of data is necessary to support various business processes.
- Data modeling provides a clear and standardized way of illustrating data structures, allowing business analysts to effectively communicate data requirements to different stakeholders.
- Business analysts can use data models to identify gaps in existing data structures and inconsistencies in business rules, hindering the successful implementation of business processes and systems.

- Data models provide a framework for organizing the data that business analysts need to analyze to make informed decisions and provide recommendations to stakeholders.
- Data models help software developers understand the data requirements and design systems that align with the business needs.
- By using data modeling, business analysts can design integrated data structures that facilitate the data exchange between different systems and applications.
- Data models provide business analysts with a structured way to assess how changes in data requirements affect business processes and ensure their continuous alignment with business objectives.

Data modeling advantages

Data modeling brings many benefits to business analysis. For instance, data models:

- help business analysts and stakeholders gain a clear understanding of the structure, relationships, and flow of data within an organization;
- act as a common language between technical and non-technical stakeholders;
- assist in defining validation rules and data constraints, ensuring data quality;
- help maintain a standardized view of data across the entire organization by requiring consistency and accuracy in data definitions;
- serve as a basis for designing databases, data storage systems, and other information systems that align with business requirements;
- aid in data-driven decision-making;
- provide the structure necessary for effective data governance;
- act as a blueprint for organizing and structuring data, facilitating data exchange between different systems.

Data modeling challenges

Although data modeling does offer numerous advantages to business analysts, it also presents certain challenges - for instance:

- The process of creating detailed data models is often very time-consuming, especially for large and complex systems.
- It's difficult to balance completeness and simplicity in data models as they can quickly become overly complex and difficult to understand, maintain, and modify.
- Existing stakeholders might resist changes proposed by new data models, especially if they disrupt established processes.
- Developing and implementing accurate and effective data models requires expertise in data modeling techniques and tools.

- Data models focus on structural aspects and might not capture all contextual or semantic nuances related to data.